ROSHAN SIDDARTHA SIVAKUMAR
615-364-5138 | roshan.siddartha.sivakumar@vanderbilt.edu | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data science graduate student with hands-on experience in machine learning, LLM applications, and production data pipelines. Built an AI chatbot presented to Nissan NA executives, engineered a BI automation system eliminating 40+ hours of weekly manual work, and led deep learning research achieving 98% accuracy on AI-generated image detection. Seeking roles where I can build scalable, real-world AI solutions.

EDUCATION

Vanderbilt University – Data Science Institute **Nashville, TN**
Master of Science in Data Science (GPA: 3.883/4) August 2024 – May 2026

- Relevant Coursework: Probability and Statistical Inference, Survey of Data Science Applications, Exploratory Data Analysis, Principles of Programming and Simulation, Data Science Rights Responsibility, Machine Learning, Data Management Systems, Deep Learning, Gen AI Models in Theory & Prac, NoSQL, Data Science Algorithms

Vellore Institute of Technology **Vellore, India**
Bachelor of Technology (GPA: 8.47/10.0) July 2019 – June 2023
Major: Mechanical Engineering

- Relevant Coursework: Problem solving and Programming, Problem Solving and Object-Oriented Programming, Statistics for Engineers, Linear Algebra and Differential Equations, Programming Data Structures and Algorithms using Python

SKILLS

Relevant Skills: Machine Learning, Deep Learning (CNNs, Transfer Learning), RAG Architectures, LLM APIs (OpenAI, Claude, Gemini), Prompt Engineering, Computer Vision (YOLOv8, Grad-CAM), ETL Pipelines, Web Scraping, Data Visualization, Feature Engineering, Statistical Modeling, SQL Querying, NoSQL Design
Related Platforms: Python (Pandas, NumPy, Scikit-learn, PyTorch, OpenCV), SQL (MySQL), MongoDB, Tableau, PowerBI, Streamlit, Flask, Amazon QuickSight, Docker, AWS, GitHub Actions, Playwright, Azure AD, Jupyter, Git

WORK EXPERIENCE

SERVPRO **Lebanon, TN**
Data Science Intern Aug 2025 – Dec 2025

- Developed a multimodal **AI agent using GPT-4 Vision** to autonomously navigate legacy enterprise portals; engineered spatial reasoning logic to visually detect and bypass dynamic pop-ups, achieving **99% scraping reliability** (Playwright).
- Architected an end-to-end **BI automation pipeline** processing **13,500+ job records nightly**; orchestrated data extraction and **Microsoft Graph API** distribution, eliminating **40+ hours** of manual weekly reporting.
- Engineered a **Python-based transformation layer (Pandas)** to process raw logs into complex financial models; programmatically generated 11 pivot tables and multi-tier AR aging reports, enabling granular tracking of revenue metrics.
- Productionalized the workflow via **GitHub Actions**, implementing **Azure AD OAuth 2.0** and secure secrets management for a **100% zero-touch**, serverless operation.

Leadership Development Program, Vanderbilt University **Nashville, TN**
AI Intern | Owen Graduate School of Management Jun 2025 – Aug 2025

- Built a secure web app to automate coach-student matching for the LDP, cutting manual work from days to under 10 minutes.
- Designed an anonymized data pipeline integrating resumes, surveys and bios to generate coach matches and rationale via OpenAI API.
- Delivered a five-column Excel output with ID-based top-2 coach matches and reasoning, enhancing transparency & scalability.
- Developed an embedded AI assistant with safety guardrails, refusal logic and prompt evaluation to support business students.

PROJECTS

Nissan Dealer Training Analytics & AI Chatbot **Nashville, TN**
Team Lead | Vanderbilt University (Capstone – Nissan North America) Aug 2025 – Dec 2025

- Analyzed 140,000+ logs to diagnose low engagement, identifying a critical gap in mobile accessibility that necessitated the move to an AI-driven, on-demand chat interface.
- Designed and deployed a survey on Nissan's training platform, collecting 800+ responses that showed 80%+ dealer interest in AI-powered product knowledge tools.
- Led development of a RAG-based chatbot prototype using Amazon QuickSight for the 2026 Nissan Leaf, integrating web-scraped specs, official documentation, and competitor data to deliver instant product comparisons.
- Presented findings and prototype to Nissan NA executives, informing a scalable chatbot solution being developed with Bethink Lab for North American dealer rollout.

AI-Generated Image Detection using Deep Learning **Nashville, TN**
Team Lead | Vanderbilt University (Deep Learning Course) Oct 2025 – Dec 2025

- Proposed and led a deep learning project to detect AI-generated images from real photographs, achieving 97.94% accuracy and 0.998 ROC-AUC on 120K images using VGG16 transfer learning.
- Implemented Grad-CAM interpretability to visualize CNN decision patterns, revealing learned distinctions between synthetic texture artifacts and natural object boundaries.

- Led architecture benchmarking across VGG16, ResNet18, and EfficientNet-B0, finding EfficientNet-B0 achieved 98.42% accuracy with 26× fewer parameters—key for resource-constrained deployment.
- Reduced training time by 40% using adaptive learning rate scheduling and early stopping, converging in 11 epochs vs. 25 planned.

Scene Reader: Real-Time Visual Accessibility via Vision-Language Models **Nashville, TN**
Team Lead | Vanderbilt University (Generative AI Course) Sep 2025 – Dec 2025

- Designed a real-time visual accessibility system for visually impaired users, benchmarking 9 transformer-based architectures (VLMs, YOLO+LLM hybrids, RAG, Chain-of-Thought) across 564 API calls and 84 local model tests.
- Built a hybrid YOLO + LLM pipeline achieving 0.54s mean latency (3× faster than pure VLM) and $0.005/query cost through semantic caching with LRU eviction and adaptive token limits.
- Architected a multi-model pipeline integrating YOLOv8, EasyOCR, and MiDaS depth estimation with complexity-based routing and parallel execution, enabling context-aware scene understanding for navigation.
- Implemented semantic caching with SHA-256 hashing, achieving 40–60% cache hit rates and 15× speedup on hits, reducing API costs for real-world deployment.

European Soccer Analytics Platform **Nashville, TN**
Solo Project | Vanderbilt University (NoSQL Course) Oct 2025 – Dec 2025

- Architected a denormalized MongoDB schema to transform 25,979 soccer matches from SQLite, embedding team/player data within documents to eliminate JOINs and reduce query latency by ~40% for analytical workloads.
- Built a Gradient Boosting classifier with 15 custom features—including rolling 5-game form tracking—to predict match outcomes across 11 European leagues, revealing that recent form (22% feature importance) outweighs static FIFA ratings.
- Developed 7 MongoDB aggregation pipelines with Flask RESTful APIs and AJAX-powered interfaces for real-time data exploration across 11,060 players and 299 teams.

Sales Playbook Optimization Dashboard **Nashville, TN**
Team Lead | Vanderbilt University (Machine Learning Course) Jan 2025 – Apr 2025

- Constructed a predictive classification pipeline using ensemble models to forecast B2B deal outcomes with >95% accuracy.
- Merged and transformed CRM data into model-ready formats with engineered behavioral and demographic features.
- Deployed a Streamlit dashboard using Docker for real-time deal scoring and win-rate monitoring.
- Enhanced pipeline robustness through iterative tuning and stratified sampling to address class imbalance.

Packaging Optimization for Automotive Supply Chain (Confidential) **Nashville, TN**
Team Lead | Vanderbilt University X Nissan (Industry Collaboration Project) Jan 2025 – Apr 2025

- Led an optimization initiative for Nissan focused on improving packaging efficiency across product families.
- Cleaned and engineered highly incomplete datasets using Python, including part grouping logic and imputation.
- Developed interactive Tableau dashboards highlighting packaging inefficiencies and standardization opportunities.
- Delivered insights projected to enable **~20% cost savings** through improved standardization and visualization.

Voter Database Design & Normalization Project **Nashville, TN**
Individual Contributor | Vanderbilt University (Database Management Systems) Jan 2025 – Apr 2025

- Structured and normalized a ~4GB MySQL database using functional dependencies and BCNF to support scalable voter registration workflows.
- Coded stored procedures, triggers, transactions, and constraints to enforce data integrity and auditability.
- Linked the backend to a PHP-based interface enabling real-time voter lookup, registration, and analytics.
- Generated SQL views and audit tables to maintain change history and promote system-wide transparency.

Economic Freedom & Societal Outcomes Analysis **Nashville, TN**
Team Lead | Vanderbilt University (Exploratory Data Analysis) Oct 2024 – Dec 2024

- Investigated global economic patterns using PCA, cluster analysis, and regression techniques on freedom, GDP, and well-being indicators.
- Segmented 160+ countries into four economic clusters using unsupervised learning to reveal structural development patterns.
- Visualized complex relationships between economic freedom, happiness, healthcare, and food affordability across top-performing nations.
- Derived policy-level insights showing that economic liberty alone does not guarantee improved societal outcomes.

ADDITIONAL

- Language Skills: English, Tamil and Telugu
- Certifications: Vanderbilt Certificate of Recognition – Nissan Packaging Optimization (Apr 2025), Programming & Data Structures in Python – NPTEL India, AI Analyst – IBM
- Interests: Story-driven gaming, WWE enthusiast, local hiking